CIS 240 LAB REPORT

Switching, Routing, Wireless Essentials / Enterprise Networks, Security, and Automation

13.5.1 Packet Tracer — WLAN Configuration

BACKGROUND	
Course:	CIS 240 - SW RT WLAN ENT & SEC
Term:	Spring 2026
Instructor:	R. Zambrano
Section Number:	21905 / 21906
Student Name:	Donaven Bruce
Student ID:	900*********
Submission Platform:	Canvas / Cisco NetAcad / Packet Tracer Activity Submission as Assigned
Report Date:	2026-04-21
Report Version:	Final
Report Title:	13.5.1 Packet Tracer - WLAN Configuration
Week / Module Reference:	Week 6 / Module 13 - WLAN Configurations
Assignment Reference:	Packet Tracer Activity 13.5.1 / WLAN Configuration
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-21
Time Spent:	Approximately 2.5 hours

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Primary host system: Windows desktop environment with Cisco Packet Tracer used for device configuration, troubleshooting, validation, and lab documentation.
- Lab platform: Cisco Packet Tracer using the provided 13.5.1 WLAN Configuration activity.
- Home wireless segment: Home Wireless Router, cable modem, Laptop, Tablet PC, Smartphone, and Internet cloud path to the web server.
- Enterprise wireless segment: RTR-1, SW1, LAP-1, WLC-1, RADIUS Server, Enterprise Admin, Wireless Host 1, and Wireless Host 2.
- Security focus areas: DHCP, WPA2-Personal, WPA2-Enterprise, RADIUS authentication, VLAN-based WLAN separation, and FlexConnect.

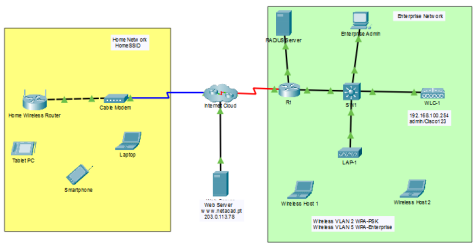


Figure 1. Initial topology state showing the home wireless network, enterprise controller environment, infrastructure devices, and required hosts before final association and verification.

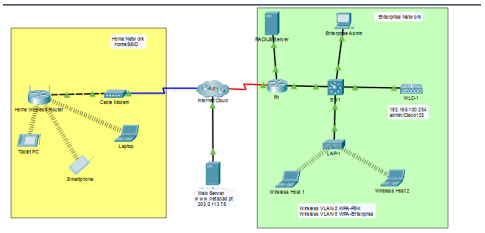


Figure 2. Final completed topology after the home WLAN, WLC interfaces, enterprise WLANs, and wireless host associations were fully validated.

Files, Media, and Resources Used:

- Packet Tracer activity file: 13.5.1 Packet Tracer - WLAN Configuration .pka.
- Lab instructions: addressing table, WLAN information table, WLC interface requirements, internal DHCP scope requirements, RADIUS details, SNMP trap receiver details, and final connectivity checks.
- Captured images used in the report: initial and final topologies, successful home-side IP assignment, successful home-side ping results, WLAN 2 interface configuration, and WLAN 5 interface configuration.
- Supporting notes: ordered configuration checklist, device-by-device association workflow, and final verification observations.

Work Objective:

- Configure a secure home wireless router using DHCP, WPA2-Personal, a defined SSID, and validated wireless client connectivity.
- Configure a wireless LAN controller with VLAN-based interfaces, an internal DHCP scope, RADIUS authentication, and SNMP settings.
- Create and validate one WPA2-PSK enterprise WLAN and one WPA2-Enterprise WLAN, then connect and test the required wireless hosts.

Learning Objectives:

- Interpret a mixed home and enterprise addressing plan from the Packet Tracer activity instructions.
- Configure a home wireless router with correct LAN addressing, DHCP behavior, DNS, and WPA2-Personal protection.
- Log in to a WLC management interface and create dynamic interfaces for VLAN-based wireless segmentation.
- Configure an internal DHCP scope, RADIUS authentication server, and SNMP trap receiver on the WLC.
- Implement both WPA2-PSK and WPA2-Enterprise WLAN security models and verify them through successful client testing.

Addressing and WLAN Plan Used:

- Home Wireless Router LAN: 192.168.6.1 /27 with DHCP pool beginning at 192.168.6.3 and maximum 20 users.
- RTR-1 subinterfaces: 192.168.2.1 /24 on G0/0/0.2, 192.168.5.1 /24 on G0/0/0.5, and 192.168.100.1 /24 on G0/0/0.100.
- WLC-1 Management: 192.168.100.254 /24; Enterprise Admin: 192.168.100.200 /24; RADIUS Server: 10.6.0.254 /24.
- WLAN 2 interface: 192.168.2.254 /24; WLAN 5 interface: 192.168.5.254 /24.
- Home WLAN: SSID HomeSSID, WPA2-Personal, passphrase Cisco123.
- Enterprise WLAN 2: SSID-2, WPA2-PSK, passphrase Cisco123.
- Enterprise WLAN 5: SSID-5, WPA2-Enterprise, username userWLAN5, password userW5pass.

PART 1 — HOME WIRELESS ROUTER CONFIGURATION

Home Router Setup Procedure:

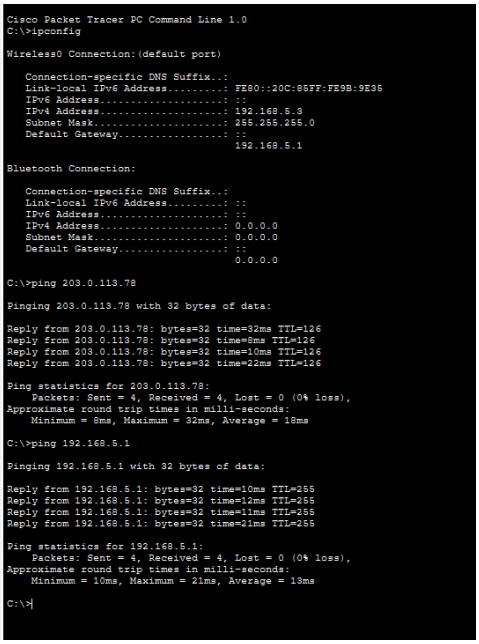
1) Opened the Home Wireless Router GUI and navigated to the Setup page.
2) Changed the Internet connection type from static addressing to Automatic Configuration - DHCP.
3) Configured the router LAN IP address as 192.168.6.1 and changed the subnet mask to 255.255.255.224.
4) Enabled the DHCP server, set the start address to host .3 of the home subnet, and set the maximum number of users to 20.
5) Configured the static DNS server as 10.100.100.252.
6) Saved the router setup and verified that the DHCP range shifted to the 192.168.6.0 /27 network.

Home Wireless and Administrative Security Procedure:

1) Opened the Wireless tab and configured the 2.4 GHz network with SSID HomeSSID.
2) Set the standard channel to 6 and kept SSID broadcast enabled so the home hosts could discover the WLAN.
3) Opened the Security tab and selected WPA2-Personal with AES encryption.
4) Configured the pre-shared key as Cisco123.
5) Opened the Administration tab and changed the router management password to Cisco123.
6) Saved the configuration and treated the home router as complete before beginning host association tests.

Home Client Association Workflow:

1) Opened the Laptop Desktop > PC Wireless utility and connected the device to HomeSSID using WPA2-Personal and passphrase Cisco123.
2) Opened the Tablet PC Config > Wireless0 interface and entered HomeSSID, WPA2-PSK, AES, and Cisco123 while leaving IP assignment on DHCP.
3) Repeated the same home-side WLAN profile on the Smartphone so all required home wireless devices were covered.
4) Used DHCP address assignment, ICMP testing, and browser access to validate that the home-side WLAN was functioning end to end.



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Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20C:85FF:FE9B:9E35
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.5.3
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.5.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 203.0.113.78

Pinging 203.0.113.78 with 32 bytes of data:

Reply from 203.0.113.78: bytes=32 time=32ms TTL=126
Reply from 203.0.113.78: bytes=32 time=8ms TTL=126
Reply from 203.0.113.78: bytes=32 time=10ms TTL=126
Reply from 203.0.113.78: bytes=32 time=22ms TTL=126

Ping statistics for 203.0.113.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 32ms, Average = 18ms

C:\>ping 192.168.5.1

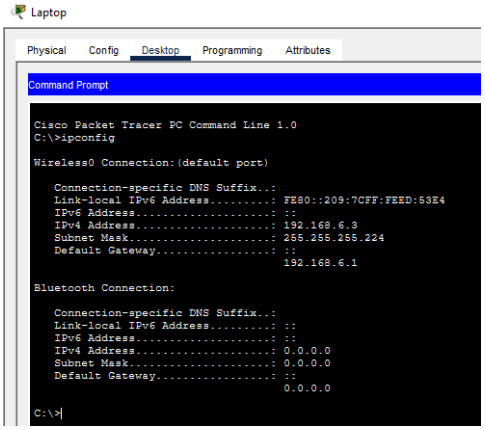
Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time=10ms TTL=255
Reply from 192.168.5.1: bytes=32 time=12ms TTL=255
Reply from 192.168.5.1: bytes=32 time=11ms TTL=255
Reply from 192.168.5.1: bytes=32 time=21ms TTL=255

Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 21ms, Average = 13ms

C:\>
```

Figure 3. Successful home-side DHCP assignment showing the Laptop receiving 192.168.6.3 /27 with default gateway 192.168.6.1.



```
Laptop
Physical   Config   Desktop   Programming   Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::209:7CFF:FEED:53E4
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.6.3
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.6.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Figure 4. Successful home-side connectivity verification showing positive ping results to the local gateway and the remote web server.

Key Home-Side Verification Commands and Outcomes:

ROUTER SETTINGS	WIRELESS SETTINGS	VERIFY
LAN IP: 192.168.6.1	SSID: HomeSSID	ipconfig
Mask: 255.255.255.224	Channel: 6	ping 192.168.6.1
DHCP Start: 192.168.6.3	Broadcast: Enabled	ping 203.0.113.78
Max Users: 20	Security: WPA2-Personal	www.netacad.pt
DNS: 10.100.100.252	AES	Admin Password: Cisco123
Internet: DHCP	Key: Cisco123	

Part 1 Result:

- Part 1 was completed successfully because the home wireless router was configured with the correct IP plan, DHCP behavior, DNS, WPA2-Personal security, and home wireless clients were able to associate and reach both the local gateway and the remote web server.

PART 2 — WLC INFRASTRUCTURE AND SERVICES CONFIGURATION

WLC Access and Interface Configuration Procedure:

1) Opened Enterprise Admin and verified that it was using the static management-side address 192.168.100.200 /24 with default gateway 192.168.100.1.
2) Tested reachability to the WLC management address 192.168.100.254 and then accessed the controller using HTTPS from the browser.
3) Logged into WLC-1 with username admin and password Cisco123.
4) Opened Controller > Interfaces and created a dynamic interface named WLAN 2 with VLAN identifier 2.
5) Configured WLAN 2 with IP address 192.168.2.254, netmask 255.255.255.0, gateway 192.168.2.1, primary DHCP server 192.168.2.1, and port 1.
6) Created a second dynamic interface named WLAN 5 with VLAN identifier 5.
7) Configured WLAN 5 with IP address 192.168.5.254, netmask 255.255.255.0, gateway 192.168.5.1, primary DHCP server 192.168.5.1, and port 1.

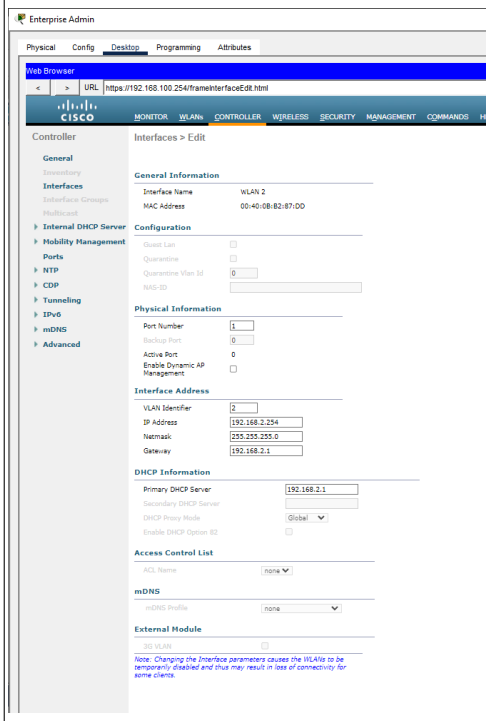


Figure 5. WLC dynamic interface configuration for WLAN 2 showing VLAN 2, interface IP 192.168.2.254, gateway 192.168.2.1, and DHCP mapping.

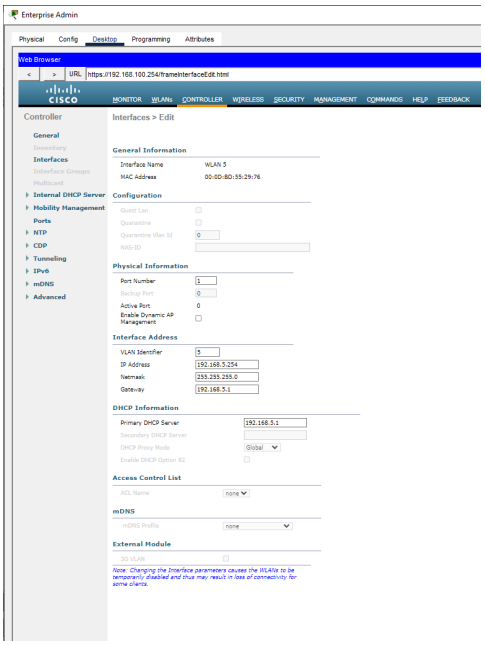


Figure 6. WLC dynamic interface configuration for WLAN 5 showing VLAN 5, interface IP 192.168.5.254, gateway 192.168.5.1, and DHCP mapping.

Internal DHCP Scope, RADIUS, and SNMP Procedure:

1) Opened Controller > Internal DHCP Server and created the management scope.
2) Configured the scope name as management with pool start 192.168.100.235 and pool end 192.168.100.245.
3) Configured the scope network as 192.168.100.0 /24 with netmask 255.255.255.0 and default router 192.168.100.1, then enabled the scope.
4) Opened Security > RADIUS > Authentication and configured server index 1 to use address 10.6.0.254 with shared secret RadiusPW.
5) Opened Management > SNMP > Trap Receivers and created an enabled receiver using community WLAN and destination 10.6.0.254.
6) Confirmed that the supporting controller-side services were in place before creating and testing the WLANs themselves.

Service Mapping and Operational Notes:

- VLAN 2 was reserved for the WPA2-PSK enterprise WLAN and VLAN 5 was reserved for the WPA2-Enterprise enterprise WLAN.
- The management network remained separate at 192.168.100.0 /24 and was used only for WLC access and related support services.
- The RADIUS server at 10.6.0.254 was required for user-based authentication on SSID-5.
- The WLC setup phase required careful tab navigation because interface creation, DHCP scope setup, authentication servers, and trap receivers were each stored in different sections of the GUI.

Part 2 Result:

- Part 2 was completed successfully because the WLC was reachable, both dynamic interfaces were addressed correctly, and the controller-side supporting services for DHCP, RADIUS, and SNMP were configured in a way that supported later WLAN association and validation.

PART 3 — ENTERPRISE WLAN CREATION AND ASSOCIATION

WLAN 2 Creation and Security Procedure:

1) Opened the WLANs page on the WLC and created Wireless VLAN 2 using SSID SSID-2 with WLAN ID 2.
2) Enabled the WLAN and changed the interface binding from the default management interface to WLAN 2.
3) Opened the Security tab and selected WPA+WPA2, then enabled WPA2 policy and AES encryption.
4) Enabled PSK and configured the pre-shared key as Cisco123.
5) Opened the Advanced tab and enabled FlexConnect Local Switching and FlexConnect Local Auth.
6) Applied the configuration and treated SSID-2 as the enterprise WPA2-PSK segment used by Wireless Host 1.

WLAN 5 Creation and Security Procedure:

1) Created Wireless VLAN 5 using SSID SSID-5 with WLAN ID 5.
2) Enabled the WLAN and bound it to interface WLAN 5.
3) Opened the Security tab, selected WPA+WPA2, enabled WPA2 policy and AES, and enabled 802.1X.
4) Disabled PSK so the WLAN would use enterprise authentication instead of shared-key authentication.
5) Opened the AAA Servers tab and assigned the configured RADIUS server entry pointing to 10.6.0.254.
6) Opened the Advanced tab and enabled FlexConnect Local Switching and FlexConnect Local Auth.

Wireless Host Association Procedure:

1) Opened Wireless Host 1 and connected it to SSID-2 using the PC Wireless utility with passphrase Cisco123.
2) Opened Wireless Host 2 and began enterprise association using the Profiles tab because direct connection to SSID-5 prompted profile creation.
3) Created a profile named SSID-5 and selected Infrastructure Mode.
4) Left network settings on DHCP to allow automatic addressing from the enterprise WLAN.
5) Configured WPA2-Enterprise using PEAP with login name userWLAN5, password userW5pass, and certificate trust-any behavior.
6) Saved the profile, returned to the Connect tab, selected SSID-5, and completed enterprise association.
7) Confirmed that the adapter became active and that the host successfully received enterprise-side addressing.

Key Enterprise-Side Settings and Workflow Summary:

SSID-2	SSID-5	HOST 2 PROFILE
Profile: Wireless VLAN 2	Profile: Wireless VLAN 5	Infrastructure Mode
SSID: SSID-2	SSID: SSID-5	DHCP
ID: 2	ID: 5	PEAP
Interface: WLAN 2	Interface: WLAN 5	userWLAN5
WPA2 + AES	WPA2 + AES + 802.1X	userW5pass
PSK: Cisco123	RADIUS: 10.6.0.254	Trust Any

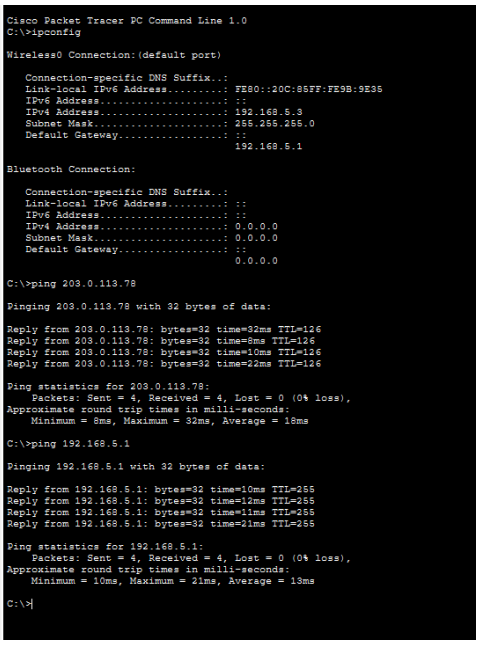
Part 3 Result:

- Part 3 was completed successfully because both enterprise WLANs were created correctly, Wireless Host 1 and Wireless Host 2 were placed on the intended SSIDs, and the enterprise-authenticated client completed the more complex profile-based WPA2-Enterprise workflow.

PART 4 — FINAL VERIFICATION, RESULTS, AND REFLECTION

Enterprise Wireless Verification Procedure:

1) Opened Wireless Host 2 command prompt and verified successful DHCP assignment with ipconfig.
2) Confirmed that Wireless Host 2 received IPv4 address 192.168.5.3 with subnet mask 255.255.255.0 and default gateway 192.168.5.1.
3) Tested local enterprise wireless reachability using ping 192.168.5.1.
4) Tested remote reachability using ping 203.0.113.78.
5) Opened the browser and confirmed that www.netacad.pt loaded successfully from the enterprise wireless host.
6) Used the same style of verification on the home-side hosts so both the consumer and enterprise portions of the activity were fully covered.



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   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.5.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

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Pinging 192.168.5.1 with 32 bytes of data:

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Approximate round trip times in milli-seconds:
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C:\>
```
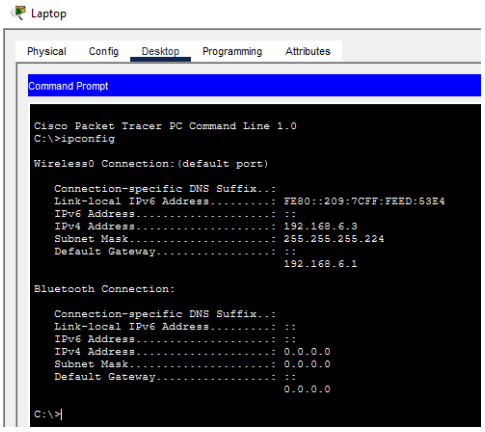

Figure 7. Host-side successful IP configuration showing correct DHCP assignment and gateway information for a validated wireless client.



```
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   Default Gateway.................: ::
                                     192.168.6.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Figure 8. Successful ping verification showing working local and remote connectivity after wireless association and addressing completed.

Completion Verification Checklist:

- Home router LAN addressing, DHCP, DNS, SSID, channel, WPA2-Personal settings, and admin password were configured correctly.
- Laptop, Tablet PC, and Smartphone were associated to the home wireless network using HomeSSID and Cisco123.
- WLC access succeeded from Enterprise Admin over the management network.
- Dynamic interfaces for WLAN 2 and WLAN 5 were created with correct VLAN IDs, IP addresses, gateways, DHCP mappings, and port numbers.
- Internal DHCP scope, RADIUS authentication server, and SNMP trap receiver were configured successfully.
- SSID-2 was configured for WPA2-PSK and SSID-5 was configured for WPA2-Enterprise with 802.1X and assigned RADIUS authentication.
- Wireless Host 2 successfully authenticated, received 192.168.5.3, and reached both the enterprise gateway and the remote web server.
- The Packet Tracer activity was saved in a stable submission-ready state.

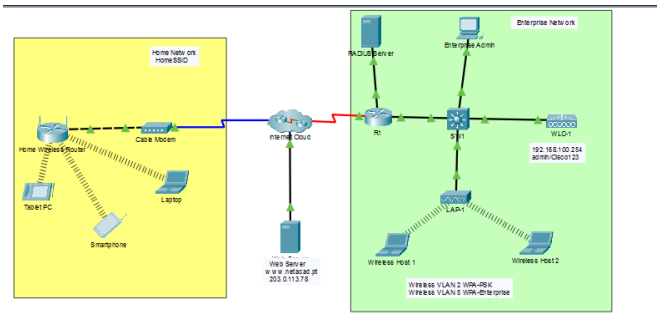

Figure 9. Final topology after full home-side and enterprise-side wireless implementation, association, and verification.

Key Findings:

1) Consumer wireless and controller-based enterprise wireless use very different workflows even when both ultimately deliver client IP connectivity.
2) Correct interface binding on the WLC is critical because associating a WLAN to the management interface instead of the intended dynamic interface will prevent proper client placement and can create misleading authentication symptoms later.
3) WPA2-Enterprise validation depends on multiple aligned elements: 802.1X, the RADIUS server entry, the client profile, and the correct user credentials.
4) Successful DHCP assignment, positive ping results, and browser access together provide stronger confirmation than any single test by itself.

Security Concepts Reinforced:

- WPA2-Personal is suitable for simpler environments but depends on a shared secret known by all users.
- WPA2-Enterprise improves control and accountability by moving authentication to a centralized RADIUS service with user-specific credentials.
- VLAN-based WLAN separation helps preserve clearer security boundaries between different wireless access policies.
- Controller-managed wireless infrastructure requires disciplined verification because one missed GUI setting can silently affect later access.

Reflection and Next Steps:

- This lab was valuable because it connected home Wi-Fi concepts to enterprise wireless controller concepts in one realistic Packet Tracer activity.
- The most important practical takeaway was seeing how enterprise wireless depends on properly aligned support services such as RADIUS, interface mapping, and client profile configuration, rather than just a shared password.
- The completed work demonstrates the ability to interpret a mixed GUI-driven networking task, troubleshoot intermediate failures, and verify secure wireless functionality from both the infrastructure side and the end-host side.
- Next step: submit the completed Packet Tracer file and final report through the assigned course platform.